YEAR 10 SCIENCE LESSON 09

How do we generate electricity?

Names: _____

Work with a partner. Visit A, B, C and D once each. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

A Moving magnet

1. Keep the magnet still. What does the meter read? _____

2. Give the magnet a small push. What does the meter do as the magnet moves?
_____

3. Talk with your partner. Why do you think that happened?

B Spin a motor

1. Keep the motor still. What does the meter read? _____

2. Spin the shaft slowly, then faster. What does the meter show?
Slow: _____ Faster: _____

3. Talk with your partner. Why do you think that happened?

C Nail and coil

1. Use the lower marked setting. How many clips can the nail lift? _____

2. Switch off. Try the higher marked setting with the same clips. How many can it lift now?
_____ Switch off again.

3. Talk with your partner. Why do you think that happened?

How do we generate electricity?

D Inside a motor

At the bench: Look inside the motor. Sketch the magnet and the coil (the loops of wire). Draw an arrow to show what turns. Then step away.

After visiting all four station types

1. What two parts can you find in both the motor and setup A?

2. What moved at A? _____ What turned at B? _____

3. Talk with your partner. Why do you think the meter changed in both tests?